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################################################################################################################
### شروع ارزیابی فقط روی مجموعه VALIDATION ###
### Dataset path: /marimo/dataset_root/val ###
### Model path: /marimo/model.pth ###
################################################################################################################
================================================================================
RGB-T ADAPTIVE TRACKER - DIAGNOSTIC EVALUATION
================================================================================
Device: NVIDIA RTX PRO 6000 Blackwell Server Edition
Selected sequences: 5

Sequences to evaluate:
   - 20190925_101846_1_4
   - 20190925_101846_1_5
   - 20190925_130434_1_2
   - 20190925_130434_1_5
   - 20190925_143900_1_1
================================================================================
[*] GPU warm-up...
[+] GPU warm-up completed.

[1/5] Sequence: 20190925_101846_1_4
[*] Sequence attributes: ['LR', 'SV', 'TC']
[*] Running Proposed Full (+ Kalman), localization_mode='modality'...

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[RESPONSE DIAGNOSTIC] Frame 50
alpha: 0.49278501135871355
RGB: {'all_scales': [(8, 6), (8, 6), (8, 6)], 'middle_peak': (8, 6), 'middle_value': 10.823227882385254}
IR: {'all_scales': [(8, 7), (8, 7), (8, 7)], 'middle_peak': (8, 7), 'middle_value': 8.39746379852295}
FUSED: {'all_scales': [(8, 6), (8, 6), (8, 6)], 'middle_peak': (8, 6), 'middle_value': 5.035785675048828}

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[RESPONSE DIAGNOSTIC] Frame 100
alpha: 0.5014290687765219
RGB: {'all_scales': [(9, 14), (9, 14), (9, 14)], 'middle_peak': (9, 14), 'middle_value': 10.7962646484375}
IR: {'all_scales': [(8, 14), (8, 14), (8, 14)], 'middle_peak': (8, 14), 'middle_value': 8.583786964416504}
FUSED: {'all_scales': [(8, 14), (8, 14), (8, 14)], 'middle_peak': (8, 14), 'middle_value': 4.928990364074707}

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[RESPONSE DIAGNOSTIC] Frame 200
alpha: 0.49463253017464764
RGB: {'all_scales': [(2, 5), (2, 6), (2, 6)], 'middle_peak': (2, 6), 'middle_value': 10.58087158203125}
```

```
IR: {'all_scales': [(3, 6), (3, 6), (4, 6)], 'middle_peak': (3, 6), 'middle_value': 8.713912963867188}
FUSED: {'all_scales': [(3, 6), (3, 6), (3, 6)], 'middle_peak': (3, 6), 'middle_value': 4.484618186950684}

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[RESPONSE DIAGNOSTIC] Frame 300
alpha: 0.5027542900172975
RGB: {'all_scales': [(2, 6), (2, 6), (2, 6)], 'middle_peak': (2, 6), 'middle_value': 10.152332305908203}
IR: {'all_scales': [(4, 6), (4, 6), (4, 7)], 'middle_peak': (4, 6), 'middle_value': 8.269043922424316}
FUSED: {'all_scales': [(3, 6), (3, 6), (3, 6)], 'middle_peak': (3, 6), 'middle_value': 4.466148376464844}

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[RESPONSE DIAGNOSTIC] Frame 400
alpha: 0.4678198178523394
RGB: {'all_scales': [(8, 5), (8, 5), (8, 5)], 'middle_peak': (8, 5), 'middle_value': 10.397171974182129}
IR: {'all_scales': [(8, 6), (8, 6), (8, 6)], 'middle_peak': (8, 6), 'middle_value': 8.770386695861816}
FUSED: {'all_scales': [(8, 5), (8, 6), (8, 6)], 'middle_peak': (8, 6), 'middle_value': 4.777841091156006}

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[RESPONSE DIAGNOSTIC] Frame 500
alpha: 0.49844761814915445
RGB: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 3.7767398357391357}
IR: {'all_scales': [(10, 11), (9, 8), (9, 9)], 'middle_peak': (9, 8), 'middle_value': 3.372182607650757}
FUSED: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 1.8103649616241455}

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[RESPONSE DIAGNOSTIC] Frame 700
alpha: 0.1643159403552888
RGB: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 3.592108964920044}
IR: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 3.4139065742492676}
FUSED: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 3.1287829875946045}

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[RESPONSE DIAGNOSTIC] Frame 900
alpha: 0.4105156789348292
RGB: {'all_scales': [(10, 10), (8, 9), (8, 9)], 'middle_peak': (8, 9), 'middle_value': 3.5716071128845215}
IR: {'all_scales': [(15, 16), (14, 16), (7, 3)], 'middle_peak': (14, 16), 'middle_value': 3.4327073097229004}
FUSED: {'all_scales': [(15, 16), (14, 16), (8, 2)], 'middle_peak': (14, 16), 'middle_value': 1.6077220439910889}

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IR: mSA=30.36% | Precision=48.20% | Success AUC=32.44% | CLE=87.97px
Visible: mSA=34.95% | Precision=44.50% | Success AUC=36.99% | CLE=200.97px
FPS=70.60
[+] Frame logs: 1000
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[2/5] Sequence: 20190925_101846_1_5
[*] Sequence attributes: ['LR', 'SV', 'TC']
[*] Running Proposed Full (+ Kalman), localization_mode='modality'...

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[RESPONSE DIAGNOSTIC] Frame 50
alpha: 0.4939986747172008
RGB: {'all_scales': [(7, 8), (7, 8), (7, 8)], 'middle_peak': (7, 8), 'middle_value': 9.699864387512207}
IR: {'all_scales': [(7, 7), (7, 7), (7, 8)], 'middle_peak': (7, 7), 'middle_value': 8.68147087097168}
FUSED: {'all_scales': [(7, 8), (7, 8), (7, 8)], 'middle_peak': (7, 8), 'middle_value': 4.795314788818359}

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[RESPONSE DIAGNOSTIC] Frame 100
alpha: 0.48568871454159457
RGB: {'all_scales': [(7, 8), (7, 8), (8, 8)], 'middle_peak': (7, 8), 'middle_value': 8.42927360534668}
IR: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 9.317495346069336}
FUSED: {'all_scales': [(7, 8), (7, 8), (8, 8)], 'middle_peak': (7, 8), 'middle_value': 4.523313045501709}

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[RESPONSE DIAGNOSTIC] Frame 200
alpha: 0.153000019888803075
RGB: {'all_scales': [(12, 10), (12, 10), (12, 9)], 'middle_peak': (12, 10), 'middle_value': 3.7623462677001953}
IR: {'all_scales': [(8, 8), (8, 8), (8, 7)], 'middle_peak': (8, 8), 'middle_value': 4.205135822296143}
FUSED: {'all_scales': [(8, 8), (8, 8), (8, 7)], 'middle_peak': (8, 8), 'middle_value': 2.8087573051452637}

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[RESPONSE DIAGNOSTIC] Frame 300
alpha: 0.18303522854009446
RGB: {'all_scales': [(2, 13), (2, 13), (2, 13)], 'middle_peak': (2, 13), 'middle_value': 3.9113962650299072}
IR: {'all_scales': [(10, 8), (11, 8), (10, 8)], 'middle_peak': (11, 8), 'middle_value': 4.2320709228515625}
FUSED: {'all_scales': [(10, 8), (10, 8), (10, 8)], 'middle_peak': (10, 8), 'middle_value': 2.912075996398926}

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[RESPONSE DIAGNOSTIC] Frame 400
alpha: 0.3265691881370397
RGB: {'all_scales': [(10, 0), (10, 0), (10, 0)], 'middle_peak': (10, 0), 'middle_value': 3.628502130508423}
IR: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 4.332316875457764}
FUSED: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 2.422612428665161}
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[RESPONSE DIAGNOSTIC] Frame 500
alpha: 0.3108872373962473
RGB: {'all_scales': [(0, 11), (1, 11), (4, 12)], 'middle_peak': (1, 11), 'middle_value': 3.966838836669922}
IR: {'all_scales': [(8, 9), (8, 9), (8, 9)], 'middle_peak': (8, 9), 'middle_value': 4.247673988342285}
FUSED: {'all_scales': [(8, 8), (8, 9), (8, 9)], 'middle_peak': (8, 9), 'middle_value': 2.4068808555603027}

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[RESPONSE DIAGNOSTIC] Frame 700
alpha: 0.7563961850420309
RGB: {'all_scales': [(9, 8), (9, 8), (9, 8)], 'middle_peak': (9, 8), 'middle_value': 4.011308193206787}
IR: {'all_scales': [(16, 11), (15, 16), (15, 16)], 'middle_peak': (15, 16), 'middle_value': 4.274420261383057}
FUSED: {'all_scales': [(9, 8), (9, 8), (9, 8)], 'middle_peak': (9, 8), 'middle_value': 2.236856460571289}

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[RESPONSE DIAGNOSTIC] Frame 900
alpha: 0.595432100465923
RGB: {'all_scales': [(8, 8), (8, 8), (0, 2)], 'middle_peak': (8, 8), 'middle_value': 3.9591176509857178}
IR: {'all_scales': [(16, 9), (15, 16), (15, 16)], 'middle_peak': (15, 16), 'middle_value': 4.268750190734863}
FUSED: {'all_scales': [(9, 8), (8, 8), (0, 16)], 'middle_peak': (8, 8), 'middle_value': 1.1940085887908936}

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IR: mSA=12.63% | Precision=17.70% | Success AUC=16.40% | CLE=223.06px
Visible: mSA=13.25% | Precision=17.80% | Success AUC=16.97% | CLE=386.22px
FPS=70.19
[+] Frame logs: 1000

[3/5] Sequence: 20190925_130434_1_2
[*] Sequence attributes: ['LR', 'OC', 'OV', 'SV', 'TC']
[*] Running Proposed Full (+ Kalman), localization_mode='modality'...

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[RESPONSE DIAGNOSTIC] Frame 50
alpha: 0.050000000371502595
RGB: {'all_scales': [(7, 5), (7, 5), (7, 5)], 'middle_peak': (7, 5), 'middle_value': 4.497361660003662}
```

```
IR: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 6.651325702667236}
FUSED: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 4.23981237411499}

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kernel_template: torch.Size([1, 256, 6, 6])
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[RESPONSE DIAGNOSTIC] Frame 100
alpha: 0.05000000000000002
RGB: {'all_scales': [(7, 5), (7, 5), (7, 5)], 'middle_peak': (7, 5), 'middle_value': 4.544538974761963}
IR: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 6.739090442657471}
FUSED: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 4.265614032745361}

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[RESPONSE DIAGNOSTIC] Frame 200
alpha: 0.05000000000000002
RGB: {'all_scales': [(7, 5), (7, 5), (7, 5)], 'middle_peak': (7, 5), 'middle_value': 4.492470741271973}
IR: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 6.703592300415039}
FUSED: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 4.27249813079834}

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[RESPONSE DIAGNOSTIC] Frame 300
alpha: 0.05000000000000002
RGB: {'all_scales': [(7, 5), (7, 5), (7, 5)], 'middle_peak': (7, 5), 'middle_value': 4.498335838317871}
IR: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 6.695776462554932}
FUSED: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 4.262816429138184}

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[RESPONSE DIAGNOSTIC] Frame 400
alpha: 0.05000000000000002
RGB: {'all_scales': [(7, 5), (7, 5), (7, 5)], 'middle_peak': (7, 5), 'middle_value': 4.541683673858643}
IR: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 6.688257217407227}
FUSED: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 4.248091697692871}

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[RESPONSE DIAGNOSTIC] Frame 500
alpha: 0.05000000000000002
RGB: {'all_scales': [(8, 4), (8, 4), (8, 4)], 'middle_peak': (8, 4), 'middle_value': 4.472160339355469}
IR: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 6.716414451599121}
FUSED: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 4.249413967132568}

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[RESPONSE DIAGNOSTIC] Frame 700
alpha: 0.050000000000001085
RGB: {'all_scales': [(8, 5), (7, 5), (7, 5)], 'middle_peak': (7, 5), 'middle_value': 4.606037139892578}
IR: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 6.689968109130859}
FUSED: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 4.201629638671875}

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===== ISM SHAPE DIAGNOSTIC =====
feat_search: torch.Size([3, 256, 22, 22])
kernel_template: torch.Size([1, 256, 6, 6])
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response_map: torch.Size([3, 1, 17, 17])
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[RESPONSE DIAGNOSTIC] Frame 900
alpha: 0.05000000000000002
RGB: {'all_scales': [(8, 4), (8, 4), (8, 4)], 'middle_peak': (8, 4), 'middle_value': 4.434736728668213}
IR: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 6.71240758895874}
FUSED: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 4.239490985870361}

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IR: mSA=0.10% | Precision=1.32% | Success AUC=4.76% | CLE=300.62px
Visible: mSA=0.10% | Precision=0.00% | Success AUC=4.76% | CLE=485.27px
FPS=69.70
[+] Frame logs: 1000

[4/5] Sequence: 20190925_130434_1_5
[*] Sequence attributes: ['FM', 'SV', 'TC']
[*] Running Proposed Full (+ Kalman), localization_mode='modality'...

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response_map: torch.Size([3, 1, 17, 17])
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===== ISM SHAPE DIAGNOSTIC =====
feat_search: torch.Size([3, 256, 22, 22])
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[RESPONSE DIAGNOSTIC] Frame 50
alpha: 0.5134173490579912
RGB: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 9.18109130859375}
IR: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 10.944853782653809}
FUSED: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 4.95048713684082}

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===== ISM SHAPE DIAGNOSTIC =====
feat_search: torch.Size([3, 256, 22, 22])
kernel_template: torch.Size([1, 256, 6, 6])
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[RESPONSE DIAGNOSTIC] Frame 100
alpha: 0.5217328518203472
RGB: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 8.874074935913086}
IR: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 10.609809875488281}
```

```
FUSED: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 4.667994499206543}

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[RESPONSE DIAGNOSTIC] Frame 200
alpha: 0.47781018456834423
RGB: {'all_scales': [(5, 8), (5, 8), (5, 8)], 'middle_peak': (5, 8), 'middle_value': 8.870935440063477}
IR: {'all_scales': [(8, 15), (8, 15), (8, 15)], 'middle_peak': (8, 15), 'middle_value': 10.460396766662598}
FUSED: {'all_scales': [(8, 16), (8, 15), (8, 15)], 'middle_peak': (8, 15), 'middle_value': 2.311230182647705}

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[RESPONSE DIAGNOSTIC] Frame 300
alpha: 0.49190569064168915
RGB: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 9.466737747192383}
IR: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 10.732636451721191}
FUSED: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 4.880072593688965}

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[RESPONSE DIAGNOSTIC] Frame 400
alpha: 0.4861561055504606
RGB: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 9.462019920349121}
IR: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 10.696159362792969}
FUSED: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 4.879650592803955}

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[RESPONSE DIAGNOSTIC] Frame 500
alpha: 0.523471973363727
RGB: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 8.888932228088379}
IR: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 10.50762939453125}
FUSED: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 4.884987831115723}

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[RESPONSE DIAGNOSTIC] Frame 700
alpha: 0.5384312805523467
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RGB: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 9.035019874572754}
IR: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 9.968311309814453}
FUSED: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 4.742775917053223}

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[RESPONSE DIAGNOSTIC] Frame 900
alpha: 0.5330690739564072
RGB: {'all_scales': [(8, 9), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 8.334573745727539}
IR: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 7.7792887687683105}
FUSED: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 4.823064804077148}

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IR: mSA=80.95% | Precision=99.70% | Success AUC=79.52% | CLE=3.19px
Visible: mSA=85.09% | Precision=99.40% | Success AUC=83.46% | CLE=6.15px
FPS=69.38
[+] Frame logs: 1000

[5/5] Sequence: 20190925_143900_1_1
[*] Sequence attributes: ['FM', 'SV', 'TC']
[*] Running Proposed Full (+ Kalman), localization_mode='modality'...

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[RESPONSE DIAGNOSTIC] Frame 50
alpha: 0.9499873508501092
RGB: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 5.351853847503662}
IR: {'all_scales': [(5, 1), (5, 1), (5, 1)], 'middle_peak': (5, 1), 'middle_value': 2.830587148666382}
FUSED: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 3.5895657539367676}

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[RESPONSE DIAGNOSTIC] Frame 100
alpha: 0.4669535606568883
RGB: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 8.081563949584961}
IR: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 8.225003242492676}
FUSED: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 4.807426452636719}

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[RESPONSE DIAGNOSTIC] Frame 200
alpha: 0.9421918595732419
RGB: {'all_scales': [(8, 9), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 4.084502220153809}
IR: {'all_scales': [(1, 0), (1, 0), (1, 0)], 'middle_peak': (1, 0), 'middle_value': 2.580202341079712}
FUSED: {'all_scales': [(8, 9), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 2.76387095451355}

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[RESPONSE DIAGNOSTIC] Frame 300
alpha: 0.7195880256883622
RGB: {'all_scales': [(8, 8), (8, 8), (7, 8)], 'middle_peak': (8, 8), 'middle_value': 3.55671763420105}
IR: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 2.8326125144958496}
FUSED: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 2.8862955570220947}
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[RESPONSE DIAGNOSTIC] Frame 400
alpha: 0.38303974859578604
RGB: {'all_scales': [(8, 8), (8, 9), (8, 8)], 'middle_peak': (8, 9), 'middle_value': 3.3105380535125732}
IR: {'all_scales': [(8, 9), (8, 9), (8, 8)], 'middle_peak': (8, 9), 'middle_value': 4.291646480560303}
FUSED: {'all_scales': [(8, 8), (8, 9), (8, 8)], 'middle_peak': (8, 9), 'middle_value': 2.80027437210083}

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[RESPONSE DIAGNOSTIC] Frame 500
alpha: 0.19425044622096413
RGB: {'all_scales': [(10, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 3.3350818157196045}
IR: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 5.045910358428955}
FUSED: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 3.1371006965637207}

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===== ISM SHAPE DIAGNOSTIC =====
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[RESPONSE DIAGNOSTIC] Frame 700
alpha: 0.18443679834997379
RGB: {'all_scales': [(8, 7), (8, 9), (8, 9)], 'middle_peak': (8, 9), 'middle_value': 3.5702884197235107}
IR: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 5.290724277496338}
FUSED: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 3.444809913635254}

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[RESPONSE DIAGNOSTIC] Frame 900
alpha: 0.14462618028887098
RGB: {'all_scales': [(8, 3), (7, 1), (7, 1)], 'middle_peak': (7, 1), 'middle_value': 3.7336695194244385}
IR: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 5.403375625610352}
FUSED: {'all_scales': [(8, 8), (8, 8), (8, 8)], 'middle_peak': (8, 8), 'middle_value': 3.0379836559295654}

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IR: mSA=5.69% | Precision=7.00% | Success AUC=10.00% | CLE=223.81px
Visible: mSA=6.63% | Precision=7.60% | Success AUC=10.90% | CLE=426.18px
FPS=72.58
[+] Frame logs: 1000
[+] Sequence metrics saved: /marimo/results_archive_val/run_20261111_211144/diagnostic_sequence_metrics.csv
[+] Frame-level CSV saved: /marimo/results_archive_val/run_20261111_211144/frame_level_diagnostics.csv
[+] Frame-level JSON saved: /marimo/results_archive_val/run_20261111_211144/frame_level_diagnostics.json
[+] ISM columns included: ism_rgb, ism_ir, ism_sim_raw_rgb, ism_sim_raw_ir
[+] Diagnostic plots saved.

================================================================================================
DIAGNOSTIC EVALUATION COMPLETED
================================================================================================
Completed sequences: 5 / 5
Output directory:
/marimo/results_archive_val/run_20261111_211144
================================================================================================

##########################################################################################################################
### ارزیابی VALIDATION با موفقیت به پایان رسید ###
##########################################################################################################################
```